Signal Processing - 1.2.30

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Question

Suppose $y_p(x) = xcos(2x)$ is a particular solution of

$$y^{''} + \alpha y = sin(2x)$$

Then the constant α equals:

1. -4
2. -2
3. 2
4. 4

Theoretical Solution

We are given a second-order linear non-homogeneous differential equation:

$$y'' + \alpha y = \sin(2x) \tag{1}$$

Given that the particular solution is $y_p(x) = x\cos(2x)$, we must determine the value of the constant α.

Step 1: Derivation of the Characteristic Equation To understand the nature of the system, we first examine the homogeneous equation $y'' + \alpha y = 0$. This represents the system's natural behavior without external forcing.

1.The Exponential Substitution We assume a solution of the form $y = e^{rx}$, where r is a constant. We test this by finding the derivatives:

- $y = e^{rx}$
- $y' = re^{rx}$
- $y'' = r^2 e^{rx}$

Theoretical Solution

Step 2: Algebraic Reduction Substituting these into the homogeneous equation:

$$r^2 e^{rx} + \alpha e^{rx} = 0$$
$$e^{rx}(r^2 + \alpha) = 0$$

Since the exponential function e^{rx} is never zero, the term in the parentheses must be zero. This gives us the **Characteristic Equation**:

$$r^2 + \alpha = 0 \tag{2}$$

The roots are $r = \pm i\sqrt{\alpha}$, which correspond to a natural oscillation frequency of $\omega_0 = \sqrt{\alpha}$.

Theoretical Solution

Mathematical Verification via Substitution We verify the constant α by substituting the given particular solution $y_p(x) = x\cos(2x)$ directly into the original non-homogeneous equation.

1.Calculus of the Particular Solution Using the product rule $\frac{d}{dx}[uv] = u'v + uv'$:

$$
\begin{aligned}
y_p(x) &= x\cos(2x) \\
y_p'(x) &= \cos(2x) - 2x\sin(2x) \\
y_p''(x) &= -2\sin(2x) - [2\sin(2x) + 4x\cos(2x)] \\
y_p''(x) &= -4\sin(2x) - 4x\cos(2x)
\end{aligned}
$$

Theoretical Solution

2.Solving for α Substitute y_p and y_p'' into $y'' + \alpha y = \sin(2x)$:

$$(-4\sin(2x) - 4x\cos(2x)) + \alpha(x\cos(2x)) = \sin(2x)$$

Group the terms involving $x\cos(2x)$:

$$-4\sin(2x) + (\alpha - 4)x\cos(2x) = \sin(2x)$$

For this equality to hold for all x, the coefficient of the $x\cos(2x)$ term must be zero:

$$\alpha - 4 = 0 \implies \alpha = 4$$

Theoretical Solution

Conceptual Analysis of Resonance: The presence of the x multiplier in $y_p(x) = x\cos(2x)$ is the mathematical signature of **Pure Resonance**. Resonance occurs when the frequency of the external driving force ($\omega = 2$ from $\sin(2x)$) perfectly matches the natural frequency of the system ($\omega_0 = \sqrt{\alpha}$). When these frequencies match, the amplitude of the system grows linearly over time, represented by the x term.

$$\text{Driving Frequency} = \text{Natural Frequency} \implies 2 = \sqrt{\alpha} \implies \alpha = 4$$

By both direct substitution and resonance theory, we find that:

$$\alpha = 4$$

The correct option is **d**.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import odeint

# Define system: y'' = sin(2x) - alpha*y
def system(state, x, alpha):
    y, v = state
    dydx = v
    dvdx = np.sin(2*x) - alpha * y
    return [dydx, dvdx]

# Timeline and start state (at rest)
x = np.linspace(0, 50, 1000)
start = [0, 0]
```

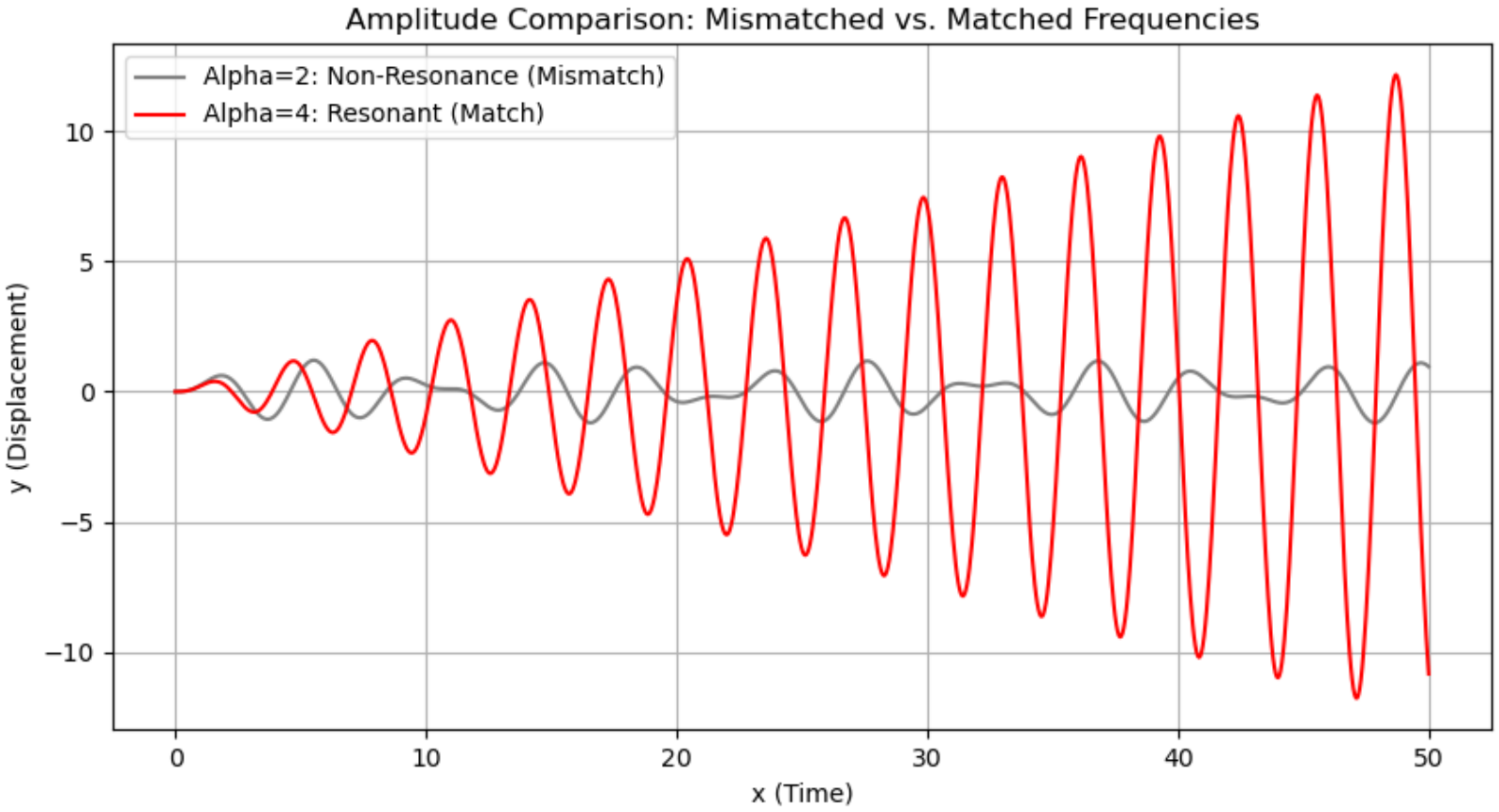
Python Code

```
# Solve for Non-resonance (Alpha=2) and Resonance (Alpha=4)
y_non = odeint(system, start, x, args=(2,))[:, 0]
y_resonant = odeint(system, start, x, args=(4,))[:, 0]

# Plotting
plt.figure(figsize=(10, 5))
plt.plot(x, y_non, label='Alpha=2: Non-Resonance (Mismatch)',
    color='gray')
plt.plot(x, y_resonant, label='Alpha=4: Resonant (Match)', color=
    'red')

plt.title('Amplitude Comparison: Mismatched vs. Matched
    Frequencies')
plt.xlabel('x (Time)')
plt.ylabel('y (Displacement)')
plt.legend()
plt.grid(True)
plt.show()
```

Plot


Amplitude Comparison: Mismatched vs. Matched Frequencies
Alpha=2: Non-Resonance (Mismatch)
Alpha=4: Resonant (Match)
y (Displacement)
x (Time)
10
5
0
−5
−10
0
10
20
30
40
50